Certifying Deep Learning Classifier Predictions with High Confidence

A PAC-Bayes-with-Backprop study: MNIST $\rightarrow$ Fashion-MNIST $\rightarrow$ CIFAR-10

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The problem

- Deep nets generalize well, but a test-set error gives *no formal guarantee* about unseen data.
- We want a risk certificate: a bound on the true error that holds with chosen confidence (e.g. 99%).
- PAC-Bayes theory gives such certificates for *probabilistic* nets (weights drawn from a learned distribution) — but for decades they were vacuous (> 1) for deep nets.

PAC-Bayes in one slide

A stochastic predictor samples weights $w \sim Q$. PAC-Bayes bounds the Gibbs risk $R(Q)$ by the empirical risk plus a complexity term:

$$R(Q) \;\lesssim\; \mathrm{kl}^{-1}\Big(\hat{R}(Q) \;\Big\|\; \tfrac{\mathrm{KL}(Q\|P)+\log(2\sqrt{n}/\delta)}{n}\Big)$$

- Tight when the posterior Q is close to the prior P (small KL).
- Non-vacuous requires (i) randomising the predictor (finite KL) and (ii) a prior near the posterior.

PAC-Bayes-with-Backprop (Pérez-Ortiz et al., 2021)

Train Q by *minimising a PAC-Bayes bound as the loss* (reparameterised, differentiable).

- **Objectives:** `fquad`, `flambda` (tight relaxations) vs `fclassic` (looser) vs `bbb` (Bayes-by-Backprop, ELBO — not a bound).
- **Priors:** data-free (`rand`) vs data-dependent (`learnt`, trained on a 50% subset).
- **Predictors:** stochastic (what the cert bounds) / posterior-mean / ensemble (what you deploy).

Setup: a difficulty ladder

- **MNIST** (reproduce Pérez-Ortiz 2021, Table 1), **Fashion-MNIST** (new extension), **CIFAR-10** (analysis).
- FCN + CNN architectures; SGD, lr 5×10^{-3}, $\sigma_0 = 0.03$, 100 epochs; risk certificates by PAC-Bayes-*kl* inversion.
- 26 runs total; every number traceable to a versioned run.

MNIST reproduction: faithful

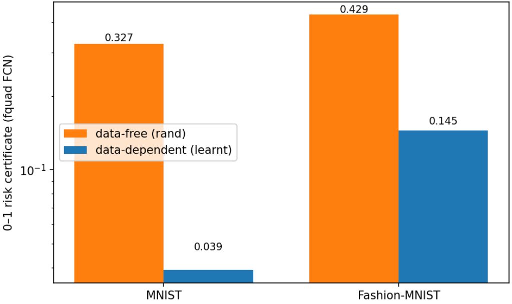
Test errors reproduced within ≤ 0.4pp of the paper; the only gap is certificate looseness from a smaller MC budget. Key pattern:

- `bbb`: best accuracy (2.0% err) but loosest cert (0.61) — it doesn't minimise a bound.
- Learnt prior tightens the `fquad` cert $\sim 8\times$: $0.327 \rightarrow 0.039$.

Fashion-MNIST extension: the method transfers

- Same qualitative findings as MNIST.
- Learnt prior tightens the `fquad` cert $\sim 3\times$: $0.429 \to 0.145$.
- Certificates stay non-vacuous on the harder task.

Figure 3. The learnt prior is the dominant lever on tightness

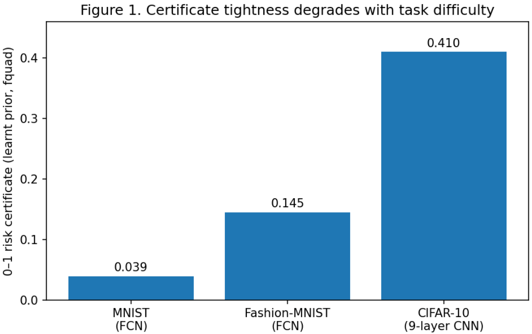


Cross-dataset: certificate vs difficulty

Learnt-prior `fquad` certificate rises monotonically with task difficulty:

$$0.039 \to 0.145 \to 0.410$$

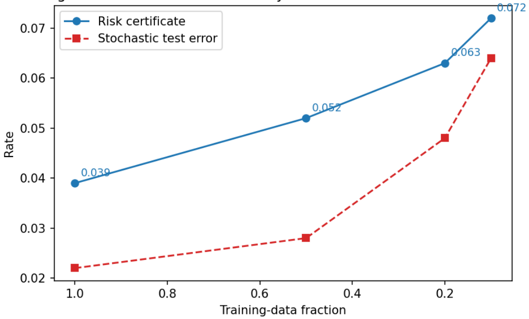
Informative on MNIST/Fashion-MNIST; approaching vacuity on CIFAR-10 — driven by irreducible error.



Small-data: bound stays non-vacuous at 10% data

Self-certified bound degrades gracefully: data $\times\frac{1}{10}$, cert only $\sim \times 2$ ($0.039 \to 0.072$, still $\ll 1$).

Figure 2. Self-certified bound stays non-vacuous down to 10% data



Takeaways

- Bound-minimising training works: competitive accuracy *and* non-vacuous certificates.
- The prior is everything: data-dependent priors are the largest lever on tightness.
- Certificates track difficulty (informative $\rightarrow$ vacuous across the ladder); the self-certified bound is data-efficient.
- Open question: a certificate *valid for the deployed (mean) predictor*, not just the stochastic one.

Thank you / Questions

Report + code + 26 runs: `pacbayes-cert`

Reference reproduced: Pérez-Ortiz et al., *JMLR* 2021 (arXiv:2007.12911).